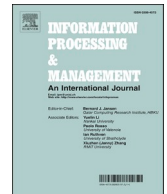




Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/infoproman

MFND-DCL: Multimodal fake news detection based on dual contrastive learning

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ARTICLE INFO

Dataset link: [Weibo \(Original data\)](#)

Keywords:

Multimodal fake news detection
 Dual contrastive learning
 Feature enhancement
 Cross-modal alignment
 Social media analysis

ABSTRACT

The rapid proliferation of multimodal fake news on social media necessitates robust detection methods, yet existing approaches often struggle to capture fine-grained cross-modal inconsistencies and discriminate semantically similar samples. To address these issues, this study proposes MFND-DCL, a Multimodal Fake News Detection framework based on Dual Contrastive Learning. At the cross-modal level, we align text and image features in a shared semantic space via contrastive loss, thereby capturing cross-modal consistency. At the inter-sample level, supervised contrastive learning is employed to optimize the feature space layout, enhancing the distinction between samples that are semantically similar but differ in veracity. Experiments conducted on the Weibo ($N = 9,527$ posts) and PHEME ($N = 5,802$ tweets) datasets demonstrate that the proposed method significantly outperforms existing baseline models in both accuracy and F1 score, achieving accuracies of 95.5% and 90.1%, respectively. Moreover, the method maintains stable performance across different language scenarios, validating the effectiveness and robustness of dual contrastive learning. This study offers new perspectives for the early identification of fake news. We release the code and data at <https://github.com/chenyujia-max/MFND-DCL>.

1. Introduction

Multimodal fake news has emerged as a critical challenge in the digital age, eroding public trust and threatening social stability (Mitra et al., 2017). However, the tendency of users to consume news via social media has also led to a significant increase in the creation and rapid spread of false information (Shrivastava et al., 2020; K. Shu et al., 2020). Fake news typically refers to deliberately fabricated misinformation intended to mislead the public (Kai Shu et al., 2017), with far-reaching consequences—it can manipulate public opinion, interfere with elections (Vosoughi et al., 2018), incite panic during public health crises, and even provoke social violence. During the COVID-19 pandemic, the dissemination of fake news contributed to vaccine hesitancy among some populations, directly hindering the implementation of containment measures (L.F. Li et al., 2020; Montagni et al., 2021; Sui et al., 2023; Tasnim et al., 2020). Meanwhile, increasingly sophisticated image forgery techniques and complex dissemination strategies have made fabricated content visually more deceptive and capable of rapid diffusion across social networks, posing ongoing challenges to traditional detection methods. Therefore, developing accurate multimodal fake news detection methods is of critical importance for safeguarding the health of the online information ecosystem and ensuring public safety.

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Received 29 December 2025; Received in revised form 21 February 2026; Accepted 7 April 2026

Available online 24 April 2026

0306-4573/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1.1. Motivation

With the widespread adoption of social media, news content increasingly appears in multimodal forms that integrate text and images (Luvembe et al., 2025; Shang et al., 2026; Wei et al., 2025; C. Yang et al., 2025). Early fake news detection research relied primarily on textual analysis, leveraging pre-trained models such as BERT and RoBERTa to extract semantic features (Devlin et al., 2019). While effective for unimodal text, these methods struggle with complex misinformation accompanied by forged or irrelevant images (Jiang et al., 2024), necessitating a multimodal approach. In recent years, multimodal detection approaches have gained traction, aiming to improve performance by fusing textual, visual, and even propagation structural features (Farhangian et al., 2024). For example, some studies employ cross-modal attention mechanisms to capture interactions between text and images (Vaswani et al., 2017), while others utilize graph neural networks to model news diffusion paths and identify anomalous spreading patterns (T. Sun et al., 2023). These methods demonstrate that integrating multimodal information can provide a more comprehensive perspective for fake news detection.

Nevertheless, a core challenge remains: forgery traces are often subtle, and falsehoods frequently arise from semantic mismatches rather than obvious tampering. For instance, in Fig. 1, a genuine photo of mixed-race child model Deniz Kane is paired with alarming text warning about child trafficking, forming a completely false narrative despite credible unimodal components. Traditional methods that rely on shallow feature concatenation or loose global alignment can easily overlook such fine-grained semantic discrepancies (Khattar et al., 2019). Moreover, semantic gaps during feature fusion and single-granularity consistency modeling limit the ability to capture complex cross-modal relationships (Shang et al., 2026), resulting in insufficient discriminative power for such deceptive samples.

Unlike traditional classification methods that depend on decision boundary separation, contrastive learning actively shapes the structure of the feature space by pulling positive samples closer and pushing negative samples apart, leading to more compact intra-class representations and more separable inter-class representations (Radford et al., 2021). This characteristic is particularly suitable for identifying highly deceptive samples (hard negatives) in fake news detection. Through cross-modal contrastive learning, models can be compelled to learn deep semantic alignment between modalities, thereby exposing text-image inconsistencies (L.Z. Wang et al., 2023). Through sample-wise contrastive learning, the global feature space layout can be further optimized, enhancing the model's ability to discriminate between semantically similar yet veracity-different samples (J.W. Li et al., 2024).

Based on the above analysis, this study proposes a Multimodal Fake News Detection method based on Dual Contrastive Learning (MFND-DCL). The framework systematically improves the quality of feature representations and discriminative ability through contrastive mechanisms at both the cross-modal and inter-sample levels. Specifically, we align multimodal features, such as text and images, in a shared semantic space, capturing cross-modal semantic consistency via cross-modal contrastive learning. Supervisory signals are also introduced to optimize the global feature space layout through sample-wise contrast. While contrastive learning itself is a well-established paradigm, the key innovation of our approach lies in the joint optimization of these two contrastive objectives within a unified framework. This joint optimization addresses a fundamental conflict in multimodal fake news detection: enforcing tight cross-modal alignment may inadvertently blur the boundaries between semantically similar real and fake samples, whereas aggressive sample separation could disrupt modality alignment. By coordinating the two objectives, our framework enables them to mutually reinforce each other—modal-level alignment provides a structured semantic space for sample-level discrimination, and sample-level separation preserves fine-grained distinctions that guide alignment away from deceptive consistency. This integrated strategy, tailored to the specific challenges of intentional mismatches and hard negatives, constitutes a coordinated optimization approach that enforces both consistency and separability at different levels of representation. Ultimately, the method achieves precise fake news detection while demonstrating strong robustness and generalization capabilities in real-world application scenarios across different cultural and linguistic contexts.

1.2. Research objectives

To address the aforementioned limitations in multimodal fake news detection, this study aims to achieve the following specific objectives:



[Year-end alert: Keep a close eye on your child!]As the year draws to a close, alarming reports of child trafficking have been circulating online, which is deeply distressing. We urge all parents to exercise extreme caution when taking children out, to prevent potential exploitation by traffickers. Please share this message with friends who have children!

Fig. 1. Example of Social Media Post.

1. To design a Dual Contrastive Learning (DCL) framework that jointly optimizes feature alignment at both cross-modal and inter-sample levels, thereby enhancing the discriminative power of multimodal representations.
2. To develop an adaptive feature fusion mechanism capable of dynamically modeling fine-grained interactions between text and image modalities, with particular attention to hard samples that exhibit semantic similarity but differ in veracity.
3. To validate the proposed model on public multilingual datasets (Weibo and PHEME) and to demonstrate its superiority over state-of-the-art baselines in terms of detection accuracy, robustness, and cross-lingual generalization.

1.3. Contributions

The main contributions of this study are as follows:

1. We propose a Dual Contrastive Learning (DCL) framework that systematically integrates sample-level and modal-level contrastive objectives within a unified multimodal detection architecture. To the best of our knowledge, this is the first work to jointly optimize inter-class separability and cross-modal alignment in a coordinated manner for fake news detection.
2. We design a Gated Cross-Attention (GCA) module that adaptively fuses text and image features by learning to gate irrelevant or inconsistent modality information. This mechanism is not a generic cross-attention variant but is specifically formulated to handle the heterogeneous noise and intentional mismatches characteristic of multimodal fake news.
3. We conduct extensive experiments on two public datasets (Weibo and PHEME), demonstrating that MFND-DCL significantly outperforms state-of-the-art methods. Beyond reporting accuracy, we also provide in-depth analyses, including feature space visualization, ablation studies, and case studies, to reveal how each component contributes to the overall performance.

The remainder of this paper is structured as follows: [Section 2](#) reviews related work. [Section 3](#) details the proposed model. [Section 4](#) describes the datasets and experimental setup. [Sections 5 and 6](#) present and discuss the experimental results. Finally, [Section 7](#) concludes the paper and outlines future research directions.

2. Related works

2.1. Research on social media-based fake news detection

Early fake news detection research primarily relied on machine learning models and manually engineered features. These hand-crafted features can generally be categorized into three types: user-related features, content-related features, and propagation network features. Researchers utilized these features to train models such as logistic regression, support vector machines, and random forests. Castillo et al. ([Castillo et al., 2011](#)) achieved preliminary fake news identification by statistically analyzing features like tweet text length, special characters, and URL count, combined with logistic regression; Gupta et al. ([Gupta et al., 2012](#)) enhanced the accuracy of event credibility assessment by fusing user social reputation features with content features. Although these methods achieved initial success on specific datasets, their performance heavily depended on the quality of feature engineering and struggled to capture deep semantic context.

The rise of deep learning technology has significantly advanced the field of fake news detection. Unimodal deep learning frameworks primarily process textual content, employing models such as CNNs, LSTMs, GRUs, and Transformers to automatically learn deep textual representations. Choudhury et al. ([Chowdhury et al., 2020](#)) constructed a sequential neural network combining four major categories of linguistic features: syntactic, sentiment, grammatical, and readability. The introduction of pre-trained language models, particularly BERT and RoBERTa, further enhanced the model's understanding of textual semantics ([Mehta et al., 2021](#); [Upadhyay & Behzadan, 2020](#)). Mehta et al. ([Mehta et al., 2021](#)) used BERT to encode statements, metadata, and justification grounds in the LIAR PLUS dataset, achieving 74% accuracy in a binary classification task. These models demonstrated performance significantly superior to traditional methods on benchmark datasets like LIAR and FakeNewsNet.

To more comprehensively address increasingly complex forms of misinformation, multimodal deep learning frameworks have become the current research mainstream ([Song et al., 2021](#)). These frameworks leverage both the textual content of news and related visual information, capturing consistency or inconsistency between modalities through multimodal information fusion ([Xue et al., 2021](#)). For instance, the SAFE model proposed by Zhou et al. ([X.Y. Zhou et al., 2020](#)) assesses the relevance between text and images by computing the similarity of their features; the MCE model by Kang et al. ([Kang et al., 2020](#)) evaluates the overall credibility of news by combining vectors from each modal component; and the BDANN model by Zhang et al. ([Zhang et al., 2020](#)) integrates BERT-extracted textual features with VGG-19-extracted image features, incorporating domain adaptation to improve generalization. Furthermore, the integration of social context information has enriched the detection dimensions, as GNNs and their variants can model complex user-news-comment relationships to capture the heterogeneous propagation patterns of fake news (Q. [Huang et al., 2019](#); [Lu & Li, 2020](#)).

In recent years, fake news detection research has exhibited an advanced trend toward multimodal fusion. The application of attention mechanisms enables models to focus on key information (T. [Chen et al., 2018](#); K. [Shu et al., 2019](#)); the introduction of knowledge graphs alleviates the fact-checking bottleneck by incorporating external factual knowledge (Cui et al., 2020; [Dun et al., 2021](#)). The development of Large Language Models (LLMs) presents new opportunities and challenges. They can address low-resource labeling problems through zero-shot or few-shot learning and leverage powerful reasoning capabilities for fact verification, but they also carry risks such as generating convincing fake news and factual hallucinations (C.Y. [Chen & Shu, 2024](#); L. [Huang et al., 2025](#)).

Moreover, LLMs find it difficult to identify fake news with fine-grained multimodal inconsistencies. Additionally, interpretability research and ethical considerations have become focal points, driving detection models to evolve from "high accuracy" toward "trustworthy and usable" (Lu & Li, 2020; J. Wang et al., 2022). Current research trends indicate that social media-based fake news detection has gradually shifted from unimodal, shallow feature modeling toward a deep model system emphasizing multimodal fusion, knowledge enhancement, interpretability, and ethics.

2.2. Contrastive learning methods and their application prospects in multimodal fake news detection

Contrastive learning is an unsupervised representation learning method whose core idea is to construct positive and negative sample pairs, pulling similar samples closer and pushing dissimilar samples apart in the feature space, thereby forcing the model to learn discriminative features and the intrinsic structure of the data (T. Chen et al., 2020; Motiian et al., 2017). This method does not rely on large amounts of labeled data and can effectively capture fine-grained differences within the data.

The core framework of contrastive learning consists of three parts: a feature encoder, positive/negative sample construction, and a contrastive loss function. The feature encoder is responsible for mapping multimodal inputs into a unified feature space. The construction of positive and negative samples is a critical step: positive samples are typically generated via intra-modal or cross-modal data augmentation of the original data, ensuring that samples remain semantically similar at their core (T. Chen et al., 2020); negative samples include dissimilar samples within a batch, cross-domain samples, or hard negatives, where the introduction of hard negatives can significantly enhance the model's ability to discern fine-grained differences (J.Y. Wu et al., 2023). Representative contrastive learning models include SimCLR in computer vision, CPC in natural language processing, and CLIP in the multimodal domain (Singh & Sharma, 2022). The core advantage of these models lies in their ability to learn high-quality feature representations without relying on labeled information, providing an effective solution for low-resource scenarios.

The application prospects of contrastive learning in multimodal fake news detection are primarily reflected in the following dimensions:

First, achieving multimodal feature alignment and fusion. One of the core challenges in multimodal fake news detection is the heterogeneity of modalities such as text and images. Contrastive learning can achieve unified spatial mapping of multimodal features through cross-modal contrastive loss, improving the effectiveness of early-fusion and late-fusion strategies (Xue et al., 2021). For example, in detecting fake news where text contradicts images, optimizing cross-modal consistency via contrastive learning can make the model more sensitive to discriminative signals indicating "textual description mismatching image content" (Song et al., 2021).

Second, addressing low-label and few-shot detection challenges. In fake news detection, many emerging events and low-resource languages lack labeled data. Contrastive learning can leverage unlabeled data to pre-train encoders, learning general feature representations, and then fine-tune with a small amount of labeled data, reducing dependency on annotated samples (J.Y. Wu et al., 2023).

Third, capturing fine-grained discriminative features. Contrastive learning reinforces the distinction between similar and dissimilar samples, prompting the model to focus on these critical differences and thereby improving detection accuracy (Singh & Sharma, 2022). For example, in health-related fake news detection, contrastive learning can help the model capture fine-grained features such as "misuse of medical terminology" and "images contradicting medical facts" (Upadhayay & Behzadan, 2020).

Fourth, enhancing model robustness and cross-domain adaptability. By exposing the model to diverse positive and negative sample pairs, contrastive learning encourages the learning of domain-invariant features, improving adaptability in cross-platform and cross-event detection. Simultaneously, constructing adversarial negative samples can enhance the model's robustness against malicious tampering (Z.X. Zhou et al., 2019).

Fifth, facilitating knowledge fusion and semantic enhancement. Contrastive learning combined with knowledge graphs can construct "news-knowledge" positive/negative sample pairs, driving the model to learn fact-consistency features and improving detection interpretability (Dun et al., 2021).

In summary, contrastive learning provides a "weakly-supervised, representation-strong, and highly robust" solution for multimodal fake news detection. It holds irreplaceable advantages, particularly in scenarios involving low resources, cross-modal fusion, and fine-grained discrimination.

3. Methodology

3.1. Problem definition

In this study, the fake news detection task is formalized as a binary classification task based on textual and visual content. The objective is to construct a model capable of fusing and reasoning across cross-modal semantics to accurately identify false information. Specifically, given a social media post $P_i = (T_i, I_i)$, where T_i denotes the textual content and I_i denotes one or more associated images, the model is required to output the predicted probability of the post belonging to either the "real" or "fake" category.

3.2. Overview framework

The multimodal fake news detection model MFND-DCL proposed in this study (as illustrated in Fig. 2) is structured around three core layers: feature extraction, interactive fusion, and contrastive optimization. First, the framework employs the CLIP model as the foundational encoder to extract semantically aligned features from text and images, respectively (Radford et al., 2021). Subsequently, a newly designed Gated Cross-Attention module enables fine-grained, adaptive interaction and fusion between textual and visual

features, thereby capturing complementary and mutually reinforcing information across modalities. Finally, the framework introduces a DCL mechanism to collaboratively optimize the feature space from two dimensions: cross-modal and inter-sample. Cross-modal contrastive learning aligns textual and visual semantics, enhancing cross-modal consistency modeling, while inter-sample contrastive learning pulls together samples of the same class and pushes apart those of different classes, improving the discriminative power of the features (Khosla et al., 2020). The entire model is jointly optimized using a classification loss and dual contrastive loss, ultimately outputting a prediction regarding the veracity of the news.

3.3. CLIP representation

In the task of multimodal fake news detection, the heterogeneity between textual and visual modalities constitutes a significant semantic gap. To establish a unified and semantically aligned foundational feature representation, this study employs the CLIP model-pre-trained on large-scale internet image-text pairs-as the cross-modal feature extractor. The core design philosophy of CLIP is to leverage contrastive learning, which brings together images and text describing the same semantic content in a shared embedding space while pushing apart unrelated image-text pairs. This enables robust cross-modal understanding and zero-shot transfer capability without the need for task-specific annotations.

Specifically, for a given image-text post $P_i = (T_i, I_i)$, where T_i denotes the textual content and I_i denotes the set of associated images, we use CLIP's visual encoder (ViT-B/16) and text encoder to extract the image and text features, respectively. Let the extracted text feature be $f_t \in \mathbb{R}^d$, and the image feature be $f_v \in \mathbb{R}^{d_v}$ (for posts containing multiple images, we take the average of their features). Through CLIP-based feature extraction, we obtain a semantically aligned multimodal feature foundation, providing high-quality input for the subsequent deep fusion and contrastive learning modules.

3.4. Gated cross-attention

Standard cross-attention mechanisms treat all cross-modal interactions uniformly, which is problematic for fake news detection where posts often contain intentionally mismatched or irrelevant images (Hao et al., 2017). Fusing such conflicting information indiscriminately introduces noise and degrades the model's ability to capture genuine semantic consistency-a critical cue for veracity assessment.

To address this, we introduce a Gated Cross-Attention module that dynamically controls cross-modal information flow based on input content (Jia et al., 2025). After computing cross-attention features, a learnable gate $g \in [0, 1]$ is generated from the concatenation of the original query and attended features, modulating the fusion to balance between preserving original modality semantics and incorporating cross-modal information. This adaptive mechanism (as illustrated in Fig. 3) suppresses misleading or irrelevant modalities while reinforcing consistent semantics, preventing the model from over-aligning mismatched pairs. Working in tandem with dual contrastive learning, this tailored design addresses the specific challenge of intentional image-text mismatches in multimodal misinformation.

Given the text feature sequence $T \in \mathbb{R}^d$ and the image feature sequence $I \in \mathbb{R}^{d_v}$, they are first mapped into Query (Q), Key (K), and Value (V) vectors via linear transformations, respectively. Specifically, for the text modality, we have:

$$Q_t = TW_t^Q, K_t = TW_t^K, V_t = TW_t^V \quad (1)$$

For the image modality, we have:

$$Q_v = IW_v^Q, K_v = IW_v^K, V_v = IW_v^V \quad (2)$$

Where $W_*^Q, W_*^K, W_*^V \in \mathbb{R}^{d \times d_k}$ are learnable projection matrices.

Subsequently, the cross-attention from text to image is computed, taking the text as the query and the image as the key-value pair:

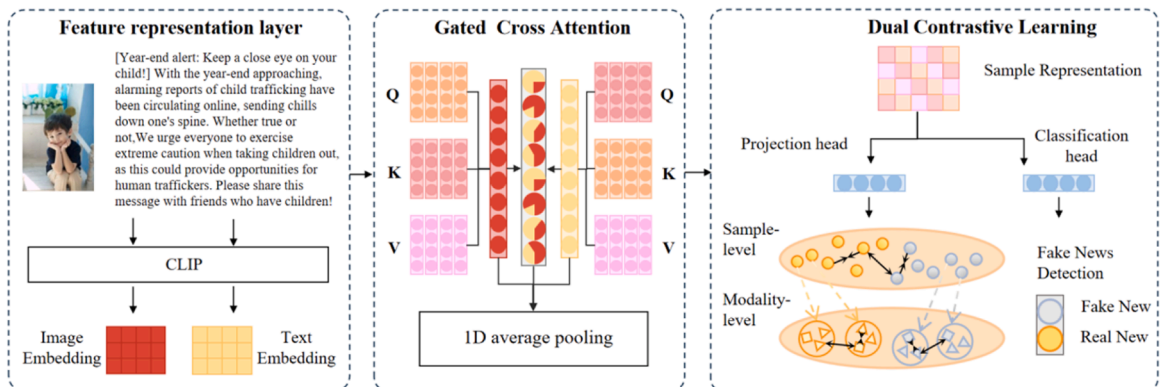


Fig. 2. MFND-DCL Framework.

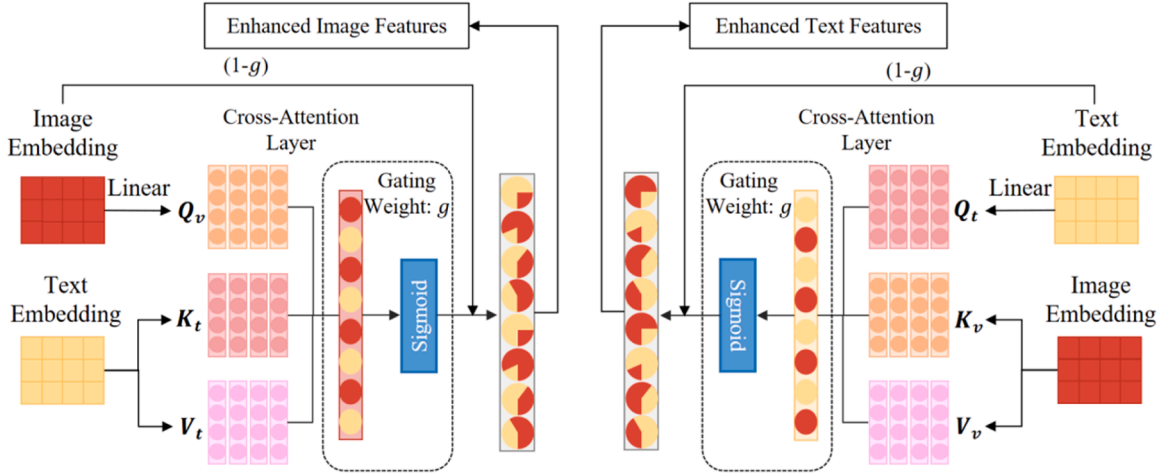


Fig. 3. Gated Cross-Attention.

$$Attention_{Q_t, K_v, V_v} = \text{softmax}\left(\frac{Q_t K_v^T}{\sqrt{d_k}}\right) V_v \quad (3)$$

Similarly, the cross-attention from image to text can be computed. To adaptively control the cross-modal information flow and suppress noise or irrelevant information, we design a gating unit. This unit takes the concatenated vector of the original query feature and the attention output feature, and generates a gating value $g \in [0, 1]$ through a non-linear mapping:

$$g = \sigma(W_g \cdot \text{Concat}(T, \text{Attention}(Q_t, K_v, V_v)) + b_g) \quad (4)$$

Where σ denotes the Sigmoid activation function, and W_g and b_g are learnable parameters. The gating value is used to modulate the fusion weight between the attention output and the original feature:

$$\tilde{T} = g \odot \text{Attention}(Q_t, K_v, V_v) + (1 - g) \odot T \quad (5)$$

Finally, the enhanced modality features are obtained through residual connection and layer normalization. This gating mechanism enables the model to dynamically adjust the intensity of cross-modal information fusion based on the input content, effectively enhancing the focus on key discriminative features while preserving the semantic integrity of each modality.

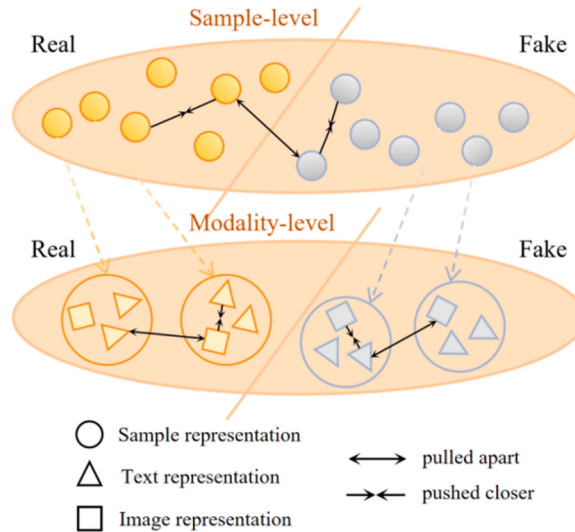


Fig. 4. Dual contrastive learning.

3.5. Dual contrastive learning

To further optimize the multimodal representation space and enhance the model's discriminative power for fake news samples, this study proposes a DCL framework (as illustrated in Fig. 4). This framework consists of two co-optimized levels: sample-level contrastive learning and modal-level contrastive learning, aiming to improve intra-class consistency, inter-class separability, and cross-modal semantic alignment, respectively.

3.5.1. Sample-level contrastive learning

Sample-level contrastive learning aims to enhance feature discriminability by pulling together samples of the same class and pushing apart samples of different classes in the projected space. This loss operates on the sample-level representation $\mathbf{z}_s \in \mathbb{R}^{d_p}$, generated from the fused features. Consider a training batch containing N samples with their corresponding labels $y_i \in \{0, 1\}$. First, L2 normalization is applied to the sample representations:

$$\hat{\mathbf{z}}_s^{(i)} = \frac{\mathbf{z}_s^{(i)}}{\|\mathbf{z}_s^{(i)}\|_2} \quad (6)$$

For an anchor sample i , its positive set $\mathcal{P}(i) = \{j | y_j = y_i, j \neq i\}$ consists of samples from the same class, while the negative set comprises all other samples from the opposite class. The similarity between samples is computed using temperature-scaled cosine similarity:

$$s_{ij} = \frac{\hat{\mathbf{z}}_s^{(i)\top} \cdot \hat{\mathbf{z}}_s^{(j)}}{\tau_s} \quad (7)$$

Where τ_s is a temperature hyperparameter that controls the sharpness of the similarity distribution. The sample-level contrastive loss $\mathcal{L}_{\text{sample}}$ is defined as:

$$\mathcal{L}_{\text{sample}} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \frac{1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp(s_{ip})}{\sum_{k=1}^N \mathbf{1}_{k \neq i} \exp(s_{ik})} \quad (8)$$

Where $\mathcal{B} = \{i | |\mathcal{P}(i)| > 0\}$ ensures that each anchor has at least one positive sample. By minimizing this loss, the model can effectively distinguish hard negative samples that are semantically similar but belong to different classes, thereby strengthening inter-class boundaries.

3.5.2. Modal-level contrastive learning

Modal-level contrastive learning aims to align the semantic representations of text and image modalities from the same post, forcing them to be closer in a shared semantic space while pushing apart modal features from different posts. This loss operates on the specific projected representations $\mathbf{z}_t$ and $\mathbf{z}_v$ from the text and image branches, respectively.

First, L2 normalization is applied to the projected representations. Within a batch, we construct a text-to-image similarity matrix $\mathbf{S}_{t \rightarrow v} \in \mathbb{R}^{N \times N}$, where each element $S_{t \rightarrow v}^{(mn)} = \frac{(\mathbf{z}_t^{(m)})^\top \mathbf{z}_v^{(n)}}{\tau_m}$. Here, τ_m is the modal-level temperature parameter. The modal-level contrastive loss consists of two symmetric InfoNCE losses:

$$\mathcal{L}_{t \rightarrow v} = -\frac{1}{N} \sum_{m=1}^N \log \frac{\exp(S_{t \rightarrow v}^{(mm)})}{\sum_{n=1}^N \exp(S_{t \rightarrow v}^{(mn)})} \quad (9)$$

$$\mathcal{L}_{v \rightarrow t} = -\frac{1}{N} \sum_{n=1}^N \log \frac{\exp(S_{v \rightarrow t}^{(nn)})}{\sum_{m=1}^N \exp(S_{v \rightarrow t}^{(mn)})} \quad (10)$$

The final modal-level contrastive loss is the average of the two:

$$\mathcal{L}_{\text{modal}} = \frac{1}{2} (\mathcal{L}_{t \rightarrow v} + \mathcal{L}_{v \rightarrow t}) \quad (11)$$

This mechanism effectively enhances cross-modal alignment, allowing textual and visual representations to complement and verify each other at the semantic level while preserving their distinct modal characteristics.

3.5.3. Overall optimization objective

The model's total loss function is obtained by taking a weighted sum of the cross-entropy loss $\mathcal{L}_{\text{CE}}$ for the classification task and the dual contrastive losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{sample}} \mathcal{L}_{\text{sample}} + \lambda_{\text{modal}} \mathcal{L}_{\text{modal}} \quad (12)$$

Where λ_{sample} and λ_{modal} are balancing hyperparameters used to adjust the influence of different loss terms on shaping the representation space. Through joint optimization, the model not only learns accurate classification decision boundaries but also acquires

feature representations with stronger discriminability and superior cross-modal alignment. This significantly enhances the robustness and generalization capability of the multimodal fake news detection system.

4. Experiment

4.1. Dataset

This section systematically details the experimental implementation. First, the basic statistical characteristics and sources of the Weibo and PHEME datasets are presented. Then, the criteria for baseline models and evaluation metrics are specified, followed by a clarification of the experimental environment configuration.

4.1.1. Weibo dataset

The Weibo dataset, constructed by Jin et al. (Jin et al., 2017), is a widely used Chinese dataset for multimodal fake news detection. It is collected from the Sina Weibo platform. The real news content originates from China's authoritative news agency, Xinhua News Agency, while the fake news is sourced from Weibo's official rumor refutation system. The dataset includes both text and image modalities, with unimodal samples filtered out to ensure multimodal integrity. To ensure data quality, near-duplicate images were removed using Locality Sensitive Hashing (LSH), and very small or excessively large images were filtered. The Weibo dataset has been validated by multiple studies as an important benchmark for multimodal fake news detection, making it suitable for research directions such as feature fusion and cross-modal alignment (M.X. Han et al., 2025; Liu et al., 2025; C. Yang et al., 2025).

Table 1 summarizes the statistics of the dataset. The training set contains 3,748 fake news items and 3,783 real news items, and the test set contains 1,000 fake news items and 996 real news items. This results in a total of 9,528 multimodal samples. The data is split strictly based on event clustering, ensuring samples from the same event do not appear across different splits, which helps evaluate the model's generalization ability.

Finally, this study adopts the typical split scheme from the dataset's original research: 80% for training and 20% for testing.

4.1.2. PHEME dataset

The PHEME dataset (Zubiaga et al., 2017) is considered a benchmark in the field of rumor analysis. It consists of tweets collected via the Twitter Streaming API during five real-world breaking news events. It contains rumor-related tweets from five major breaking news events, covering the Ferguson unrest, Ottawa shootings, Sydney siege, Charlie Hebdo shooting, and Germanwings Flight 9525 crash. These events span multiple domains, providing an important template for studying the mechanisms of misinformation dissemination in critical scenarios. The tweets were annotated by journalists, who labeled each tweet based on whether the information was verified at the time of posting, adhering to the established definition of rumors as unverified claims. As shown in Table 2, the dataset comprises 5,802 tweets, each explicitly labeled as a rumor or non-rumor, offering valuable data support for analyzing rumor propagation patterns during major events. The data were sampled based on a retweet threshold (≥ 100 retweets) to focus on high-impact posts.

This study adopts the typical split scheme from previous research (L.W. Wu et al., 2024) on this dataset: 70% for training, 10% for validation, and 20% for testing.

4.2. Baselines

In the interest of a fair and direct comparison, we perform a literature-based comparison with established baselines. The choice of baseline algorithms differs between datasets due to the distinct characteristics and prior evaluation practices. This involves using the performance metrics reported in the original papers for these methods, as they were evaluated on the same dataset and with the same data segmentation as our work.

4.2.1. Weibo dataset

To evaluate our method, we compare it with recent multimodal fake news detection approaches:

MCAN (Y. Wu et al., 2021): Uses a multi-layer co-attention mechanism to interact spatial and frequency features from images and text.

IIMR (Singhal et al., 2022): Employs product-based fusion and a weight decay loss to reduce the effect of weak modalities.

CAFE (Y.X. Chen et al., 2022): Projects features into a common space, measures cross-modal uncertainty via KL divergence, and adjusts modality contributions dynamically.

Table 1
Statistics of the Weibo Dataset.

Dataset	Split	Label	Number
Weibo	Train	Fake news	3748
		Real news	3783
	Test	Fake news	1000
		Real news	996
	Total	-	9527

Table 2
Statistics of the PHEME Dataset.

Event	Rumours	Non-rumours	Total
Charlie Hebdo	458 (22.0%)	1,621 (78.0%)	2,079
Ferguson	284 (24.8%)	859 (75.2%)	1,143
Germanwings Crash	238 (50.7%)	231 (49.3%)	469
Ottawa Shooting	470 (52.8%)	420 (47.2%)	890
Sydney Siege	522 (42.8%)	699 (57.2%)	1221
Total	1972 (34.0%)	3830 (66.0%)	5802

FNR (Ghorbanpour et al., 2023): Uses supervised contrastive learning to model text-image similarity via inner products of feature vectors.

MPFN (Jing et al., 2023): Integrates text and image frequency features at multiple levels and fuses them with a feature mixer.

CMCN (H. Han et al., 2023): Fuses text and visual objects hierarchically and applies modular cross-attention for cascaded text feature extraction.

CSFND (Peng et al., 2024): Combines unsupervised local context features with global semantics to separate similar real and fake news.

CLIP-FND (J.Z. Xu et al., 2024): Encodes text and image with CLIP, fuses them via attention and bilinear pooling, then classifies.

MRAN (H.Y. Yang et al., 2024): Uses a relation-aware attention network to compute intra- and cross-modal similarities and generates high-order fusion features.

CP-FEND (Y. Xu et al., 2024): Simulates human reasoning by guiding LLMs through a "checking-reasoning-judging" chain for better accuracy and interpretability.

PSATEL (Leite et al., 2025): Extracts 19 credibility signals from news via prompted LLMs and combines them for zero-shot authenticity prediction.

EEEMDF (Liu et al., 2025): Expands text with external entities, uses fine- and coarse-grained interaction modules, and dynamically fuses features with a routing mechanism.

EC-Fake (M.X. Han et al., 2025): Augments features using LLM-based external knowledge and contrastive learning; decouples image foreground/background to guide multi-level fusion.

4.2.2. PHEME dataset

We also compare with unimodal textual models and earlier multimodal methods:

BERT (Devlin et al., 2019): A transformer-based language model fine-tuned for rumor detection.

TextGCN (Yao et al., 2019): Treats documents as nodes in a graph to capture global word co-occurrence.

CompareNet (Hu et al., 2021): Compares textual entities with knowledge base entries via a graph network.

SpotFake (Singhal et al., 2019): Combines text features from a language model and image features from VGG-19.

SpotFake+ (Singhal et al., 2020): An enhanced version of SpotFake that extracts textual features using the XLNet model.

SAFE (X.Y. Zhou et al., 2020): A similarity-aware approach focused on studying the multimodal information of news articles.

HMCAN (Qian et al., 2021): Achieves fake news detection by jointly modeling multimodal contextual features and the hierarchical semantics of text.

KDCN (Sun et al., 2023): Checks consistency between text and images, and between content and background knowledge.

LLM-only (Yan et al., 2025): An approach where results are produced solely by a large language model following prompt instructions.

M2ERD (Yan et al., 2025): A novel framework for accurate emergency rumor detection by collaboratively leveraging large and small language models (LLMs and SLMs) and deeply integrating multimodal features.

4.3. Evaluation metrics

In this study, to comprehensively and objectively evaluate the performance of the fake news detection model, we employ Accuracy, Precision, Recall, and the F1-score as evaluation metrics.

The formulas are as follows:

$$Precision = \frac{TP}{TP + FP} \quad (13)$$

$$Recall = \frac{TP}{TP + FN} \quad (14)$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

$$F1 = \frac{2 \times Precision}{Precision + Recall} \quad (16)$$

Here, TP (True Positives) denotes the number of fake news correctly identified as fake, FP (False Positives) denotes the number of real news incorrectly identified as fake, FN (False Negatives) denotes the number of fake news incorrectly identified as real, and TN (True Negatives) denotes the number of real news correctly identified as real.

4.4. Experiment setup

To ensure fair model comparison and reproducibility of results, this study adopts a unified yet moderately adjusted training strategy across both datasets. All experiments are implemented using the PyTorch deep learning framework on an NVIDIA A100-PCIe 40GB GPU. Consistency in stochastic processes is ensured by fixing the global random seed to 129.

The model is trained using the Adam optimizer, with gradient clipping applied to stabilize the training process. The weighting coefficients for $\mathcal{L}_{CE}$, $\mathcal{L}_{sample}$, and $\mathcal{L}_{model}$ are set to 1.0, 0.5, and 0.3, respectively, and the temperature parameter is set to 0.1. For both datasets, the model is trained for a maximum of 100 epochs, and an early stopping mechanism is employed to prevent overfitting; training is terminated when the validation F1-score ceases to improve.

Considering the characteristics of each dataset, specific hyperparameters are configured as follows: for the Weibo dataset, which contains complete multimodal information, the batch size is set to 128, the initial learning rate is set to 1e-4, and the classifier dropout rate is set to 0.3. For the PHEME dataset, which exhibits more significant class imbalance, a larger batch size of 256 and a slightly lower learning rate of 6e-5 are used. Furthermore, class weights of [1.0, 2.4] are introduced into the cross-entropy loss to mitigate the imbalance issue.

5. Experiments analysis

5.1. Main results

5.1.1. Comparison results

To evaluate model performance, we conducted comprehensive experiments on the Weibo and PHEME datasets and compared our approach with existing representative methods. The results show that our model achieves leading or highly competitive outcomes on both tasks, validating its effectiveness.

As shown in Table 3, on the Weibo dataset, our model achieved optimal performance with an accuracy of 0.955, representing a significant improvement of 2.4% over the best baseline model, EC-Fake. In terms of F1-scores for fake news and real news, we also attained 0.957 and 0.953, respectively, surpassing all baseline models across the board. Compared with earlier fusion-based models such as MCAN and CMMCN, which rely on static or loosely coupled attention mechanisms, MFND-DCL explicitly enforces both cross-modal alignment and inter-sample discrimination through its DCL framework. While attention mechanisms enhance interaction between modalities, they do not structurally regulate the embedding space. In contrast, the DCL objective encourages intra-class compactness and inter-class separability, leading to clearer decision boundaries and reduced ambiguity in semantically similar samples.

Furthermore, compared with recent LLM-enhanced methods such as EC-Fake and EEEMDF, which incorporate external knowledge or entity expansion, MFND-DCL achieves superior performance without relying on external knowledge bases during inference. This suggests that the coordinated optimization of feature fusion (via Gated Cross-Attention) and embedding geometry (via dual contrastive constraints) provides a more stable and internally consistent representation structure. The notable improvement in the real news F1-score indicates that the model effectively suppresses spurious cross-modal correlations that often lead to false positive predictions.

As shown in Table 4, on the PHEME dataset, our model also achieves excellent performance, achieving an accuracy of 0.901, which is representing a significant improvement of 2.4% over the best baseline model (M2ERD). Notably, our model excelled in real news identification (as illustrated in Table 5), attaining an F1-score of 0.924. Given that PHEME is relatively text-dominant and exhibits class

Table 3

Results of comparison between MFND-DCL and other fake news detection methods on Weibo datasets.

Dataset	Method	Accuracy	Fake News		Real News			
			Precision	Recall	F1-score	Precision	Recall	F1-score
Weibo	MCAN	0.899	0.913	0.889	0.901	0.884	0.909	0.897
	IIMR	0.900	0.878	0.817	0.846	0.884	0.826	0.854
	CAFE	0.840	0.855	0.83	0.842	0.825	0.851	0.837
	FNR	0.879	0.870	0.890	0.880	0.880	0.870	0.880
	MPFN	0.838	0.857	0.894	0.889	0.873	0.863	0.876
	CMMCN	0.873	0.848	0.838	0.843	0.889	0.897	0.893
	CSFND	0.895	0.899	0.895	0.897	0.892	0.896	0.894
	CLIP-FND	0.890	0.879	0.907	0.893	0.904	0.876	0.889
	MRAN	0.903	0.904	0.908	0.906	0.897	0.892	0.894
	CP-FEND (LLM-based)	0.868	0.895	0.834	0.863	0.844	0.902	0.872
	PASTEL (LLM-based)	0.688	0.684	0.674	0.679	0.692	0.702	0.697
	EEEMDF	0.931	0.941	0.920	0.930	0.921	0.942	0.931
	EC-Fake (LLM-based)	0.931	0.932	0.979	0.955	0.906	0.967	0.936
	Ours	0.955	0.954	0.959	0.957	0.956	0.951	0.953

imbalance, this improvement highlights the robustness of the contrastive framework under modality-asymmetric conditions. In such scenarios, modal-level contrastive learning mitigates over-reliance on a single modality, while sample-level contrastive loss reinforces stable shared patterns among veracity-consistent samples. As a result, MFND-DCL maintains discriminative power even when visual signals are sparse or weak.

Overall, the superiority of MFND-DCL can be attributed not merely to architectural complexity, but to the coordinated design of gated cross-modal interaction and dual-level contrastive regularization. The former enhances fine-grained semantic fusion, while the latter reshapes the representation space to enlarge class margins and strengthen cross-modal consistency. These complementary mechanisms jointly explain the consistent performance gains observed across datasets with different linguistic, structural, and modality characteristics.

5.1.2. Complexity analysis

We analyze the computational complexity of MFND-DCL from both theoretical and empirical perspectives. The model consists of three main components: a frozen CLIP encoder used only for feature extraction in a separate preprocessing stage, a trainable GCA module, and dual contrastive learning losses. During training, the pre-extracted features are loaded directly, so the CLIP forward pass does not contribute to the reported training time.

(1) Theoretical complexity: The GCA module adds ~ 0.8 GFLOPs with $O(L_t^2 \cdot d)$ attention cost, where L_t is the text sequence length and $d = 512$. The contrastive losses compute pairwise similarities within a batch, leading to $O(B^2 \cdot d_p)$ complexity, where B is batch size and $d_p = 128$ is the projection dimension. This quadratic term grows with B , but for our settings ($B = 128$ for Weibo, $B = 256$ for PHEME) it accounts for less than 5% of total training time.

(2) Empirical results: On an NVIDIA A100 GPU with early stopping applied, the training time for Weibo is 106.7 seconds (2.7 s/epoch) with 1.46 M trainable parameters, and for PHEME is 71.0 seconds (0.9 s/epoch) with 0.94 M trainable parameters.

These measurements confirm that despite the quadratic dependency on batch size, the overall training time remains modest and well within the capacity of modern hardware, demonstrating the practical scalability of MFND-DCL.

5.2. Ablation study

To assess the contribution of each core module, we conducted ablation studies on both datasets, evaluating three progressively enhanced variants: 1) w/o GCA DCL: A classification model that uses only CLIP for feature extraction and fuses them via simple pooling and concatenation; 2) w/o DCL: This model builds upon the Baseline by incorporating the Gated Cross-Attention (GCA) module; 3) Ours (Full Model): The complete model, which adds the DCL module on top of the GCA-enhanced framework.

As shown in Table 5 and Fig. 5, on the Weibo dataset, each module brings clear gains. The w/o GCA DCL model achieves an accuracy of 0.9270. After integrating the GCA module, the model's accuracy increased to 0.9386. The F1-scores for fake and real news also improved from 0.9281 and 0.9257 to 0.9390 and 0.9381, respectively. This strongly validates that the GCA module, through dynamic and adaptive cross-modal interaction, effectively uncovers complementary and mutually corroborative relationships between text and images, thereby enhancing the quality of feature fusion. Ultimately, the full model incorporating the DCL module achieved optimal performance. Its accuracy further increased to 0.9549, with F1-scores for fake and real news reaching 0.9565 and 0.9533, respectively. This clearly demonstrates that the DCL module, via collaborative contrastive optimization at both sample and modality levels, effectively widens class boundaries and strengthens cross-modal alignment. Consequently, it further refines the overall representation space structure built upon the already powerful fused features, constituting a critical component for the model's superior performance.

On the PHEME dataset, the contribution trends of each module align with those on the Weibo dataset, though the degree of impact differs. The w/o GCA DCL model achieved an accuracy of 0.8623. Introducing the GCA module yielded a modest performance gain.

Table 4

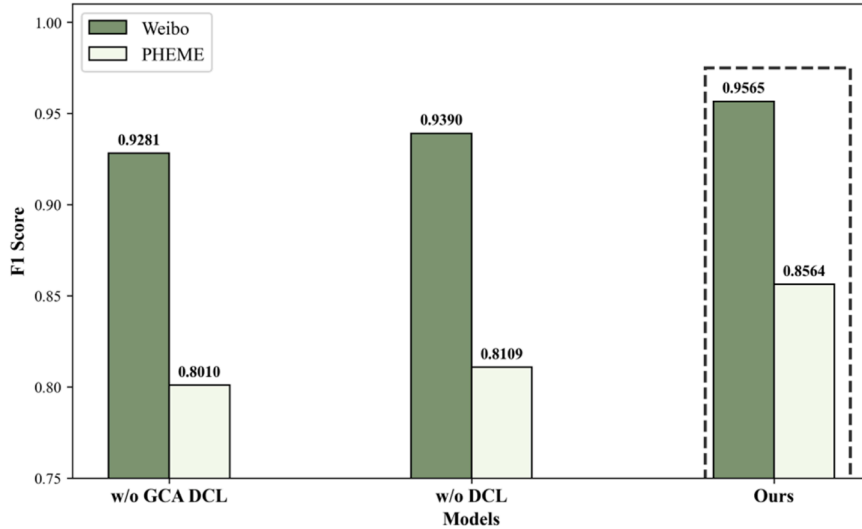
Results of comparison between MFND-DCL and other fake news detection methods on PHEME datasets.

Dataset	Method	Accuracy	Precision	Recall	F1-score
PHEME	BERT	0.826	0.789	0.786	0.787
	TextGCN	0.817	0.818	0.817	0.817
	CompareNet	0.765	0.765	0.765	0.765
	SpotFake	0.823	0.817	0.823	0.820
	SpotFake+	0.814	0.811	0.814	0.812
	SAFE	0.846	0.839	0.842	0.841
	HMCAN	0.851	0.842	0.843	0.842
	KDCN	0.833	0.800	0.804	0.802
	LLM-only	0.654	0.447	0.480	0.463
	(LLM-based)				
	M2ERD	0.877	0.862	0.861	0.861
	(LLM-based)				
	Ours	0.901	0.846	0.867	0.856

Table 5

Results of comparison between MFND-DCL and its variant models on the two datasets.

Dataset	Method	Accuracy	Fake News		Real News			
			Precision	Recall	F1-score	Precision	Recall	F1-score
Weibo	w/o GCA DCL	0.9270	0.9427	0.9140	0.9281	0.9112	0.9408	0.9257
	w/o DCL	0.9386	0.9625	0.9167	0.9390	0.9154	0.9619	0.9381
	Ours	0.9549	0.9539	0.9590	0.9565	0.9560	0.9506	0.9533
PHEME	w/o GCA DCL	0.8623	0.7902	0.8120	0.8010	0.9012	0.8883	0.8947
	w/o DCL	0.8640	0.7715	0.8546	0.8109	0.9202	0.8688	0.8938
	Ours	0.9008	0.8460	0.8672	0.8564	0.9303	0.9182	0.9242

**Fig. 5.** Results of comparison between MFND-DCL and its variant models on the two datasets.

This suggests that in the PHEME dataset, which is text-dominant with relatively sparse visual information, the benefit from cross-modal interaction is somewhat limited. However, incorporating the DCL module led to a significant performance leap, substantially boosting the accuracy to 0.9008. It is particularly noteworthy that the improvement in recognizing real news is especially prominent after introducing the DCL module, as its F1-score increased from approximately 0.8938 to 0.9242. This underscores the importance of contrastive learning in imbalanced, modality-asymmetric data.

In conclusion, the ablation studies systematically verify the effectiveness and necessity of our two proposed core modules: GCA and DCL. The GCA module serves as the foundation for enhancing multimodal fusion quality, while the DCL module is the key innovation that further optimizes the feature space, driving the model's performance to its optimum.

5.3. Effect of temperature parameter on model performance

Within the DCL framework, the temperature parameter τ serves as a crucial hyperparameter that regulates the distribution pattern of sample similarities in the embedding space. According to information theory principles, τ scales the cosine similarity, thereby controlling the relative weighting of positive and negative sample pairs in the contrastive loss.

Table 6

Effect of Temperature Parameter on Model Performance Based on Weibo Dataset.

Temperature	Accuracy	Precision	Recall	F1-score
0.05	0.9481	0.9558	0.9431	0.9494
0.1	0.9447	0.9447	0.9484	0.9465
0.15	0.9413	0.9385	0.9484	0.9434
0.2	0.9549	0.9539	0.9590	0.9565
0.25	0.9509	0.9385	0.9683	0.9531
0.3	0.9440	0.9446	0.9471	0.9458
0.35	0.9474	0.9485	0.9497	0.9491
0.4	0.9474	0.9369	0.9630	0.9498
0.45	0.9447	0.9412	0.9524	0.9467
0.5	0.9454	0.9413	0.9537	0.9474

As shown in Table 6 and Fig. 6, we tested τ from 0.05 to 0.5. Performance peaked at $\tau = 0.2$, achieving an F1 score of 0.9565, accuracy of 0.9549, and recall of 0.9590. This moderate value prevents training instability from overly small τ and avoids blurred class distinctions from overly large τ .

Notably, raising τ from 0.05 to 0.2 yielded significant gains, indicating that appropriate similarity smoothing aids generalization. The decline beyond $\tau = 0.2$ confirms τ 's role in balancing discriminative accuracy and robustness.

5.4. Effect of learning rate on model performance

As shown in Table 7 and Fig. 7, the learning rate critically affects model training. Experiments show an inverted U-shaped relationship between performance and learning rate. At $1e-5$, the F1-score is 0.9465 and accuracy is 0.9447, indicating slow convergence.

Performance improves as the learning rate increases, peaking at $1e-4$ with an F1-score of 0.9565, accuracy of 0.9549, and recall of 0.9590. This rate balances training efficiency and stability.

Further increases to $2e-4$ and $5e-4$ cause performance to decline (F1-scores of 0.9533 and 0.9519), as overly large update steps can cause oscillation and miss fine-grained features needed for fake news detection.

5.5. Effect of dual contrastive learning on sample separation

To verify the enhancement effect of the DCL module on the discriminative ability between positive and negative samples in the fake news detection task, we compared the differences in sample representation distributions before and after introducing this mechanism into the model on the Weibo dataset. The sample-level feature representations were visualized by reducing their dimensionality to a three-dimensional space using Principal Component Analysis (PCA). As shown in Fig. 8 (a), before the introduction of contrastive learning, fake and real news samples heavily overlapped, with a silhouette coefficient of only 0.0225, indicating poor class separation and ambiguous boundaries.

With the introduction of the DCL module and as training progressed, the sample distribution structure underwent significant changes. After 15 training epochs, the silhouette coefficient increased to 0.6338, and a trend of separation began to emerge in the feature space. After continuing training until the 25th epoch, the silhouette coefficient reached 0.6476. At this stage, the distribution of real news samples exhibited a noticeably clustered state, while fake news samples also formed relatively independent clusters, resulting in a clear boundary region between the two classes.

This evolutionary process fully demonstrates that the DCL mechanism effectively restructures the feature embedding space through the dual constraints of label-supervised contrastive loss and cross-modal consistency loss. Specifically, this mechanism encourages samples of the same class to be drawn closer together in the embedding space, while simultaneously pushing samples from different classes further apart. For semantically similar, hard-to-distinguish samples, DCL strengthens the model's ability to extract key discriminative features through cross-modal consistency constraints, thereby significantly improving the model's generalization performance.

5.6. Case study

To validate the practical efficacy of the proposed DCL model in the fake news detection task, we selected six typical fake news cases from the Weibo and PHEME datasets for analysis, with specific results presented in Fig. 9.

The experimental results show that the model accurately classified all cases, with the confidence scores exhibiting a differentiated distribution. Case 1 involves fake news about child safety scams. The model correctly identified it with a confidence score close to 1.00, indicating its strong ability to discern false information that exploits social hotspots to deliberately create anxiety. Case 2 concerns health-related fake news about bacteria on soda cans. The model assigned a confidence score, reflecting its reliability in detecting false

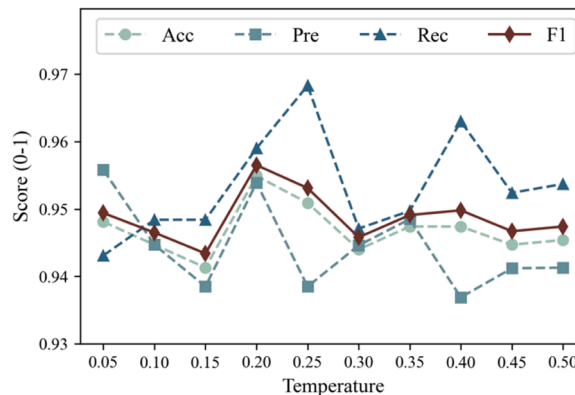
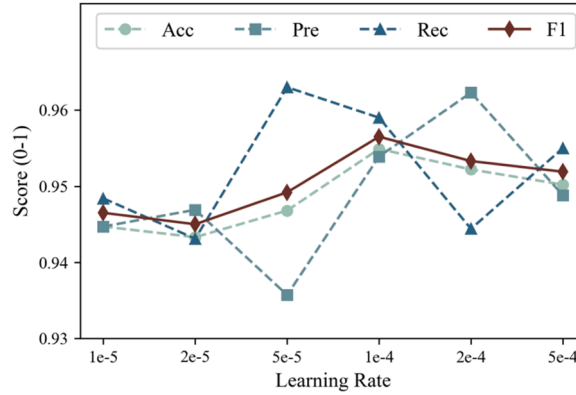


Fig. 6. Effect of Temperature Parameter on Model Performance Based on Weibo Dataset.

Table 7

Effect of Learning Rate on Model Performance Based on Weibo Dataset.

Learning Rate	Accuracy	Precision	Recall	F1-score
1e-5	0.9447	0.9447	0.9484	0.9465
2e-5	0.9433	0.9469	0.9431	0.9450
5e-5	0.9468	0.9357	0.9630	0.9492
1e-4	0.9549	0.9539	0.9590	0.9565
2e-4	0.9522	0.9623	0.9444	0.9533
5e-4	0.9502	0.9488	0.9550	0.9519

**Fig. 7.** Effect of Learning Rate on Model Performance Based on Weibo Dataset.

information related to basic scientific knowledge.

It is noteworthy that although Case 3 and Case 4 both belong to the category of social fake news, the model's confidence levels differed: 0.99 for Case 3 versus 0.97 for Case 4. This disparity reveals the model's varying sensitivity to different types of social fake news—it is more certain when judging obviously illogical fake announcements, while it remains relatively cautious with narratives involving complex social ethics.

In cross-lingual scenarios, Case 5 and Case 6 pertain to the Charlie Hebdo terrorist attack. The model made correct judgments with moderate confidence scores of 0.78 and 0.85, respectively. This reflects the adaptability of the DCL mechanism in multilingual environments. However, the complexity of international events leads to relatively conservative confidence scores, which aligns with the judgment logic in real-world application scenarios.

Further analysis indicates that the model's high-confidence judgments often correspond to two types of features: first, textual content containing obvious logical contradictions or factual errors; second, content containing vocabulary that artificially evokes exaggerated emotions. Cases with relatively lower confidence typically involve complex social events requiring verification by external knowledge. This characteristic of confidence score distribution closely aligns with the cognitive patterns of human experts when discerning fake news.

6. Discussion

This study makes several distinct theoretical and practical contributions to the field of multimodal fake news detection, advancing beyond existing approaches in the following ways.

6.1. Theoretical implication

Unlike prior methods that often treat multimodal fusion as a simple concatenation or loosely aligned aggregation, MFND-DCL introduces a dual-level contrastive learning framework that jointly optimizes both cross-modal alignment and inter-sample discrimination in a unified representation space. This provides a novel theoretical perspective on how to structurally regularize multimodal feature learning, ensuring that semantically consistent and veracity-aligned samples are pulled together while inconsistent or deceptive pairs are pushed apart. The proposed gated cross-attention mechanism further offers a dynamic and interpretable fusion strategy, which adaptively modulates inter-modal information flow based on input content, rather than relying on fixed or heuristic fusion rules. These mechanisms collectively advance the theoretical understanding of how to model fine-grained semantic mismatches and hard negative samples in multimodal misinformation.

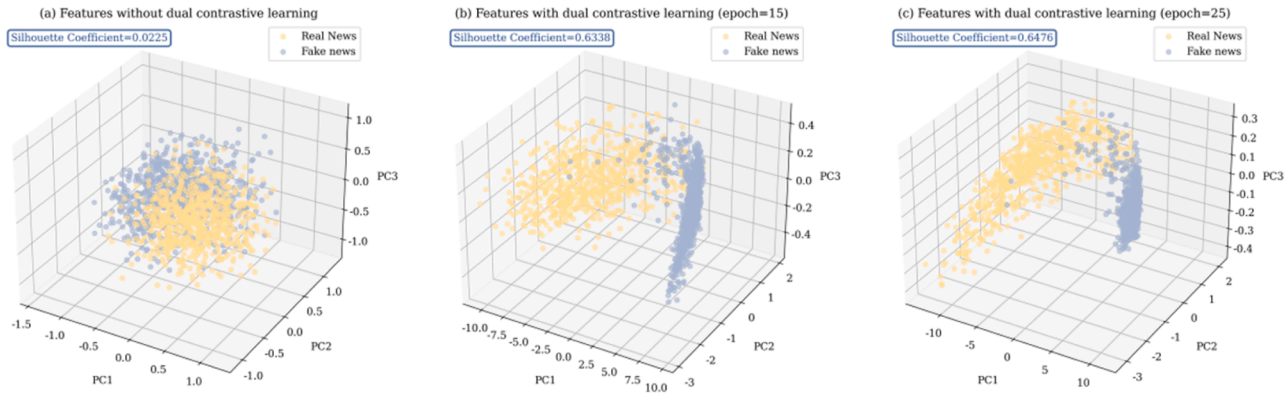


Fig. 8. Visualization of sample representations on the Weibo dataset before (a) and after (b) applying DCL.

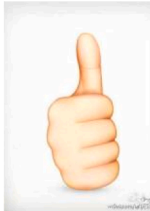
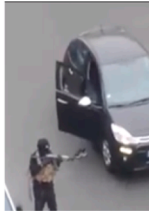
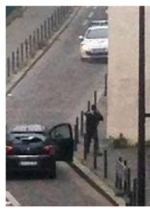
ID	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
Image						
Text	Attention, moms! Scammers are luring parents with children to pose for photos, causing them to feel dizzy when holding the camera. Then the kids get kidnapped.	A woman drank a can of beverage and was hospitalized, leaving the world. Tests confirmed that the can was contaminated with Toxoplasma gondii. The can was not cleaned during transportation.	Great news! Selling dog meat is illegal. The State Food and Drug Administration will start accepting reports of dog meat restaurants on November 1.	Dr. Li Qian assisted a pregnant woman during childbirth but was later sued for 'illegal medical practice.' Does this mean doctors lose their identity outside the hospital?	Witness to #CharlieHebdo attack claims gunmen told him it was "Al-Qaeda in Yemen"	MORE PHOTOS: Armed gunmen fire at police near
True label	Fake	Fake	Fake	Fake	Fake	Fake
Prediction	Fake(p=0.99)	Fake(p=1.00)	Fake(p=0.99)	Fake(p=0.97)	Fake(p=0.78)	Fake(p=0.85)

Fig. 9. Case study for six inference cases.

6.2. Practical implication

From an applied perspective, MFND-DCL demonstrates strong performance across two linguistically and culturally distinct datasets (Weibo and PHEME), achieving state-of-the-art accuracy and F1 scores. This demonstrates its potential for real-world deployment on multilingual social media platforms, where fake news appears in diverse forms and contexts. The model's robustness to hyperparameter variations and its ability to clearly separate real and fake news in the feature space suggest that it can reduce manual moderation efforts and improve early detection reliability. Moreover, the case study in [section 5.6](#) illustrates how the model maintains high confidence in clear-cut cases while remaining cautious in complex, context-dependent scenarios—a behavior aligned with human expert judgment, thus supporting human-AI collaborative verification workflows.

To assess the real-world applicability of MFND-DCL, we evaluate its deployment potential from three perspectives: inference efficiency, integration feasibility, and market relevance. The model delivers efficient inference on a single NVIDIA A100 GPU, achieving per-sample processing speeds that support near-real-time content moderation on social media platforms. The core modules are implemented in PyTorch and can be exported to ONNX or TensorRT for further acceleration. We also note that the model does not rely on external knowledge bases or LLM APIs during inference, making it lightweight and cost-effective for large-scale deployment. Given the increasing regulatory pressure on platforms to curb misinformation, MFND-DCL offers a practical solution that balances accuracy, efficiency, and interpretability.

6.3. Differentiation from existing work

While recent models such as EEEMDF and CLIP-FND have incorporated external knowledge, hierarchical interaction, or pre-trained vision-language encoders, they often lack a coordinated dual-level contrastive objective that simultaneously enforces intra-class compactness, inter-class separation, and cross-modal consistency. MFND-DCL moves beyond single-level contrastive or attention-based fusion by integrating both sample- and modal-level contrastive losses with a gated cross-attention module, enabling more precise capture of subtle inconsistencies between text and image. This results in significantly improved performance on hard samples where semantics are similar but veracity differs, a challenge not systematically addressed in prior works.

7. Conclusions

In this paper, we address the challenges of multimodal fake news detection on social media by proposing a detection framework based on DCL. This framework achieves fine-grained interaction and fusion between textual and visual modalities through a Gated Cross-Attention mechanism. It further incorporates both cross-modal and inter-sample contrastive learning to collaboratively optimize the feature representation space, thereby enhancing the model's ability to discriminate highly deceptive fake news.

Experiments on two public datasets (Weibo and PHEME) show that our method outperforms existing mainstream approaches in multiple metrics, including accuracy and F1-score, demonstrating strong detection performance and cross-dataset generalization. Ablation studies further validate the effectiveness of each module, namely the Gated Cross-Attention and the DCL components. Furthermore, sensitivity analysis shows the model's robustness within a reasonable hyperparameter range, and feature visualization results confirm that DCL significantly improves intra-class compactness and inter-class separation.

Despite the promising results achieved, this study has certain limitations. First, the MFND-DCL approach is relatively sensitive to hyperparameters, such as the temperature parameter; future work could investigate adaptive tuning strategies. Second, while effective in aligning consistent semantics, modal-level contrastive learning may inadvertently reduce the salience of cross-modal inconsistency signals. Third, the current method does not explicitly model the temporal dynamics and contextual evolution of news propagation; subsequent research could incorporate sequential information for more comprehensive detection. Additionally, there is room for further optimization of the model's computational efficiency, especially when processing scenarios with multiple images or long texts. Future work may explore integrating external knowledge, extending the framework to video and other multimodal data, and improving the model's interpretability and practical deployment efficiency.

CRedit authorship contribution statement

Minghui Qian: Project administration, Funding acquisition. **Shulin Guo:** Writing – review & editing. **Yujia Chen:** Writing – original draft, Visualization, Software. **Chi Wang:** Validation, Investigation. **Zirui Ye:** Formal analysis.

Declaration of competing interest

None

Acknowledgment

This research was funded by the Beijing Social Science Fund Project under the project "Research on Full-cycle Deduction and Evaluation of Ideological and Political Education in Universities Based on Generative Agents" (Grant No. 25GJA010) and the Key Research Project on Ideological and Political Work under the same project (Grant No. BJSZ2025ZD03). The views expressed in this work represent the opinions of the authors and may not necessarily be shared or endorsed by their respective employers and/or

sponsors. This manuscript is the authors' original work and has not been published nor has it been submitted simultaneously elsewhere. All authors have checked the manuscript and have agreed to the submission.

Data availability

Data will be made available on request.

[Weibo \(Original data\)](#) (google)

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